

A Practical Guide to Data Quality: Problems, Detection, and Strategy

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Abstract

This paper provides a practical guide for data practitioners to understand, identify, and address data quality issues. We present a comprehensive taxonomy of 23 common data quality problems and a catalogue of 22 detection methods, organized into a strategic framework. This work aims to equip developers with a structured approach to implementing robust data validation strategies, analogous to software testing, to improve the reliability of data-intensive systems.

Taxonomy of Data Quality Problems

- Content Quality (Accuracy & Integrity):
Wrong data, Inconsistency, Misspellings, Outliers.
- Completeness (Sufficiency of Information):
Missing Data, Semi-empty Tuples, Insufficient Metadata.
- Representational & Structural Quality:
Non-standardized Data, Misfielded Values, Wrong Categorical Data.
- Timeliness & Relevance (Fitness for Context):
Outdated Temporal Data, Distribution Shift.

Strategic Detection

1. Rule-Based & Constraint Enforcement

(Schema, Not-Null, Regex, Triggers)

2. Statistical & ML Analysis

(Outliers, Distribution Shift, Plausibility)

3. Data Comparison & Standards

(Reference Data, Fuzzy Matching, Dictionaries)

4. Human & Manual Review

(Domain Expertise, Complex Cases)

Case Study: Public Health Dataset

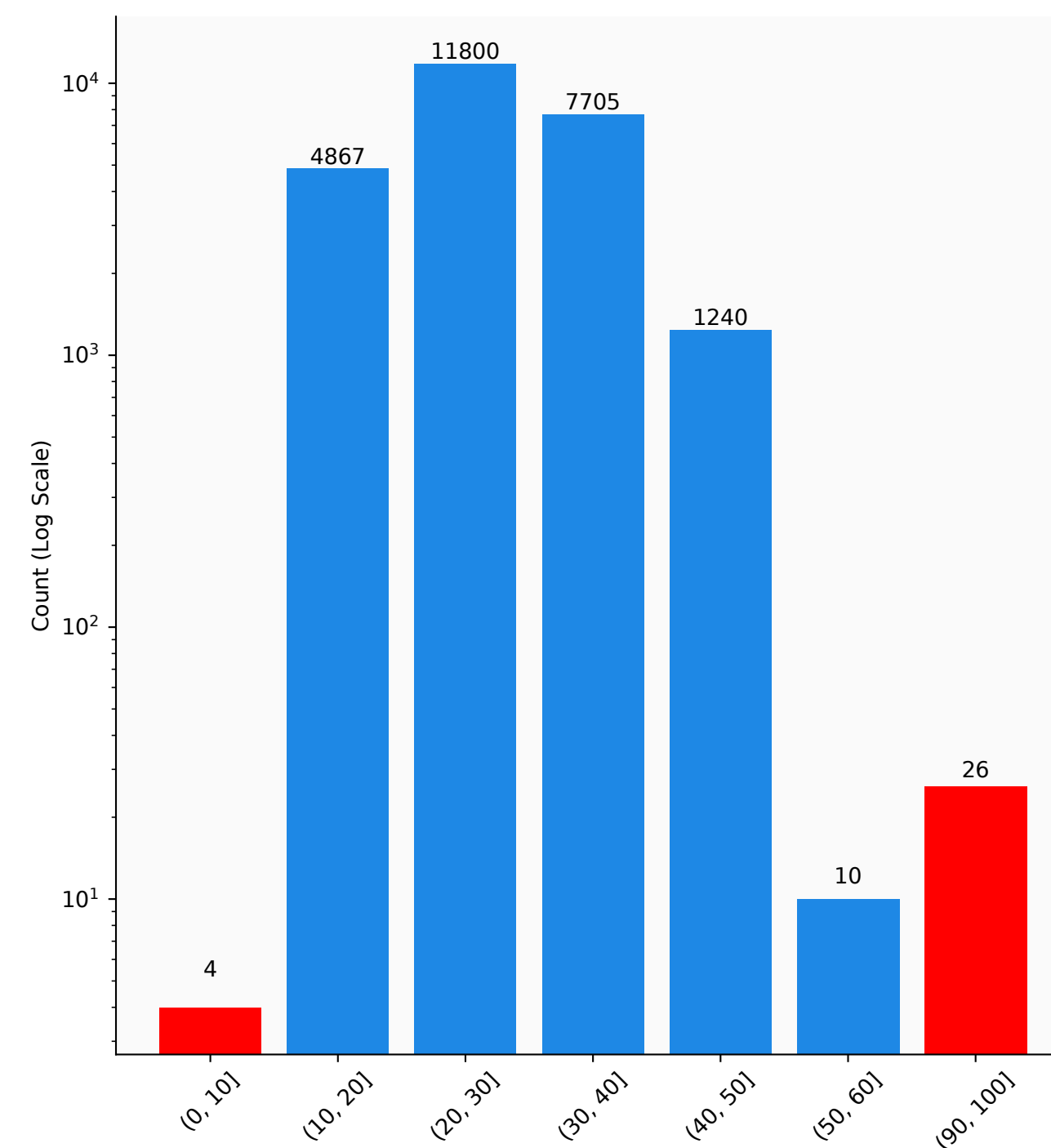
Analysis of Brazil's Mortality Information System (SIM) reveals common data quality issues.

Example 1: Inconsistency (Rule-Based Validation)

```
# Filter for records where a person
# born male is recorded as pregnant
inconsistent_records = df[
    (df.GRAVidez == 1) & (df.SEX0 == 1)
]
```

Example 2: Plausibility (Statistical Analysis)

Distribution of Mother's Age highlights extreme outliers.



Conclusions & Takeaway

Data quality assurance must be a first-class citizen in the development of data systems. By systematically identifying failure modes and applying a targeted validation strategy from automated rules to statistical checks teams can build more robust, reliable, and trustworthy data assets.